Examiner only

10. *(a)* Nitrogen can react with hydrogen to form ammonia.

$$N_2(g) \quad + \quad 3H_2(g) \quad \rightleftharpoons \quad 2NH_3(g)$$

A mixture of N_2 and H_2 is left to react at a certain temperature, until it reaches equilibrium. The equilibrium mixture has the following composition.

N_2	$1.16\,mol\,dm^{-3}$
H_2	$1.60\,mol\,dm^{-3}$
NH_3	$0.752\,mol\,dm^{-3}$

(i) A student said that the equilibrium must lie to the left because the concentrations of nitrogen and hydrogen are greater than that of ammonia. Is he correct?

Justify your answer by calculating a value for K_c for this equilibrium.
Give the unit for K_c. [4]

K_c = ...

Unit ....

....

....

(ii) When the temperature is increased the equilibrium yield of NH_3 decreases. The student said that the reaction is endothermic. Is he correct?

Justify your answer by using Le Chatelier's principle. [2]

....

....

....

....

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(b) Ammonia can be used as part of the nitrophosphate process to produce the fertiliser diammonium hydrogenphosphate (DAP) which has the formula $(NH_4)_2HPO_4$.

$$Ca(NO_3)_2 + 4H_3PO_4 + 8NH_3 \longrightarrow CaHPO_4 + 2NH_4NO_3 + 3(NH_4)_2HPO_4$$

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Calculate the maximum mass of DAP, in kg, that could be made from 1.00 tonne of ammonia. [3]

Maximum mass = ... kg

(c) Calculate the volume, in cm^3, that 2.54×10^{-3} mol of nitrogen occupies at a temperature of 120 °C and a pressure of 101 kPa. [4]

Volume = ... cm^3

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